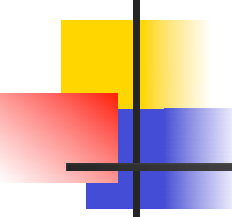
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Lecture 10

Chapter 2. Basic Structures

2.3 Functions

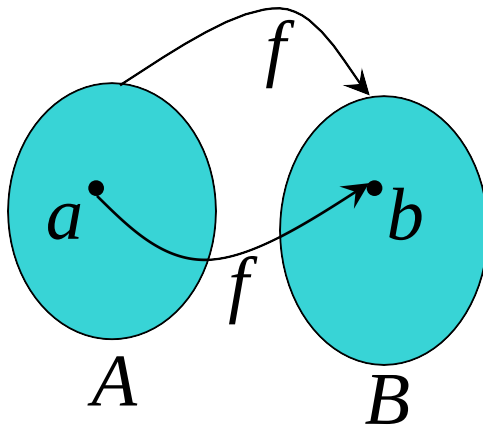
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2.3 Functions

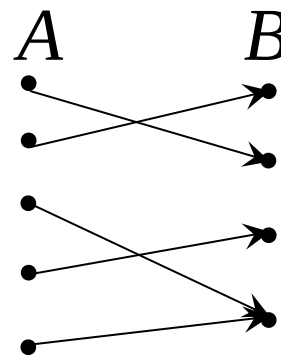
- From calculus, you are familiar with the concept of a real-valued function f , which assigns to each number $x \in \mathbf{R}$ a value $y = f(x)$, where $y \in \mathbf{R}$.
- But, the notion of a function can also be naturally generalized to the concept of assigning elements of *any* set to elements of *any* set. (Also known as a *map*.)

Function: Formal Definition

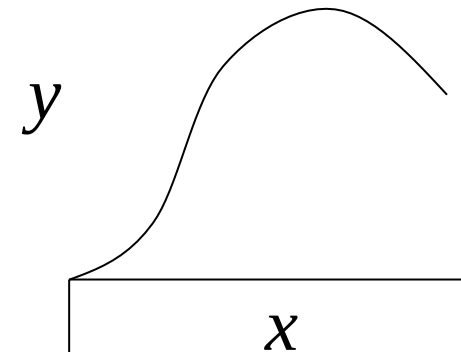
- For any sets A and B , we say that a **function** (or “**mapping**”) f from A to B ($f : A \rightarrow B$) is a particular assignment of **exactly one element** $f(x) \in B$ to **each element** $x \in A$.
- Functions can be represented graphically in several ways:



Like Venn diagrams



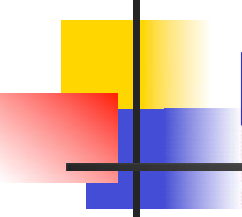
Bipartite Graph



Plot

Some Function Terminology

- If it is written that $f : A \rightarrow B$, and $f(a) = b$ (where $a \in A$ and $b \in B$), then we say:
 - A is the **domain** of f
 - B is the **codomain** of f
 - b is the **image** of a under f
 - a can not have more than 1 image
 - a is a **pre-image** of b under f
 - b may have more than 1 pre-image
 - The **range** $R \subseteq B$ of f is $R = \{b \mid \exists a f(a) = b\}$

A decorative graphic consisting of overlapping yellow, red, and blue squares with a black crosshair.

Range versus Codomain

- The range of a function might *not* be its whole codomain.
- The codomain is the set that the function is *declared* to map all domain values into.
- The range is the *particular* set of values in the codomain that the function *actually* maps elements of the domain to.

Range vs. Codomain: Example

- Suppose I declare that: “ f is a function mapping students in this class to the set of grades $\{A, B, C, D, F\}$.”
- At this point, you know f ’s codomain is: $\{A, B, C, D, F\}$, and its range is unknown!
- Suppose the grades turn out all As and Bs.
- Then the range of f is $\{A, B\}$, but its codomain is still $\{A, B, C, D, F\}$!

Function Operators

- $+$, $\times$ (“plus”, “times”) are binary operators over $\mathbf{R}$. (Normal addition & multiplication.)
- Therefore, we can also add and multiply two *real-valued functions* $f, g: \mathbf{R} \rightarrow \mathbf{R}$:
 - $(f + g): \mathbf{R} \rightarrow \mathbf{R}$, where $(f + g)(x) = f(x) + g(x)$
 - $(fg): \mathbf{R} \rightarrow \mathbf{R}$, where $(fg)(x) = f(x)g(x)$
- Example 6:

Let f and g be functions from $\mathbf{R}$ to $\mathbf{R}$ such that $f(x) = x^2$ and $g(x) = x - x^2$.
What are the functions $f + g$ and fg ?

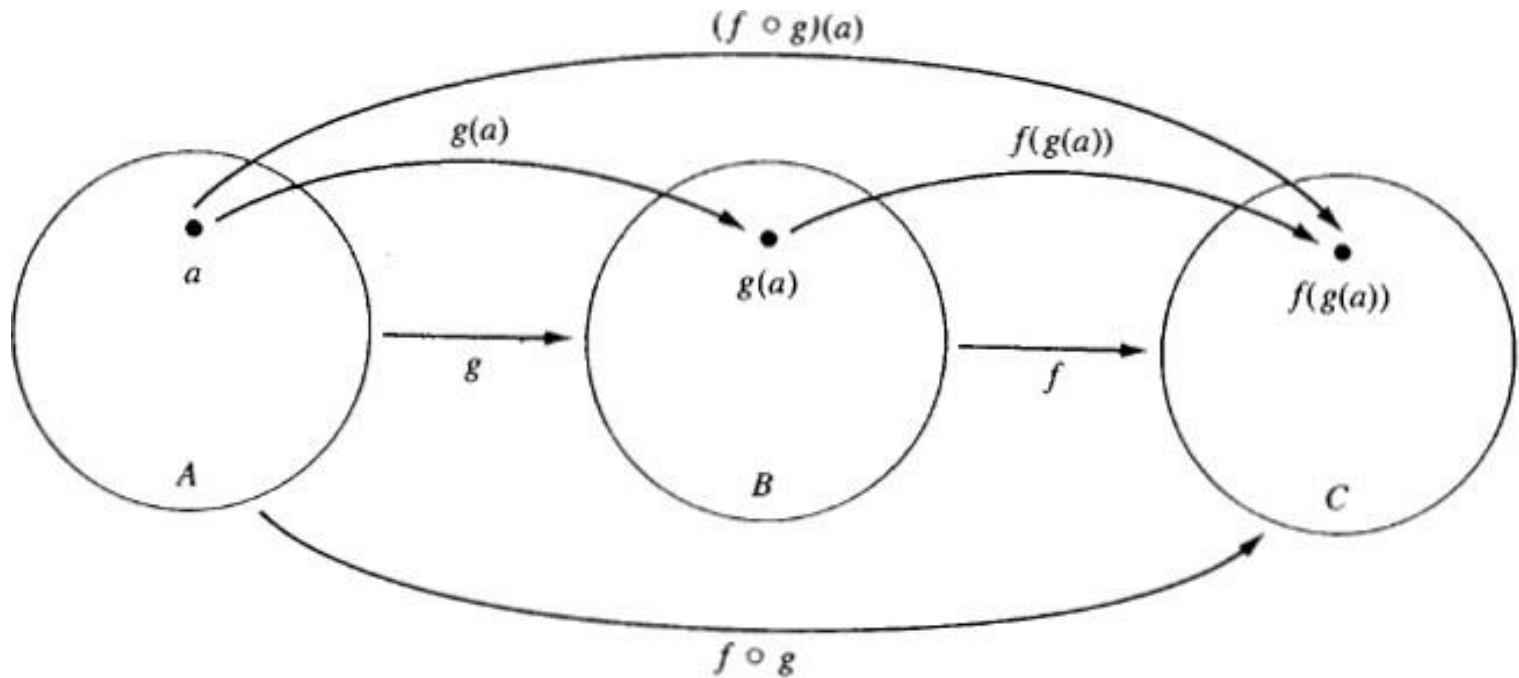
Function Composition Operator

Note the match here. It's necessary!

- For functions $g: A \rightarrow B$ and $f: B \rightarrow C$, there is a special operator called **compose** (“ $\circ$ ”).
 - It composes (creates) a new function from f and g by applying f to the result of applying g .
 - We say $(f \circ g): A \rightarrow C$, where $(f \circ g)(a) = f(g(a))$.
 - Note: $f \circ g$ cannot be defined unless range of g is a subset of the domain of f .
 - Note $g(a) \in B$, so $f(g(a))$ is defined and $\in C$.
 - Note that $\circ$ is non-commuting. (Like Cartesian $\times$, but unlike $+$, $\wedge$, $\cup$) (Generally, $f \circ g \neq g \circ f$.)

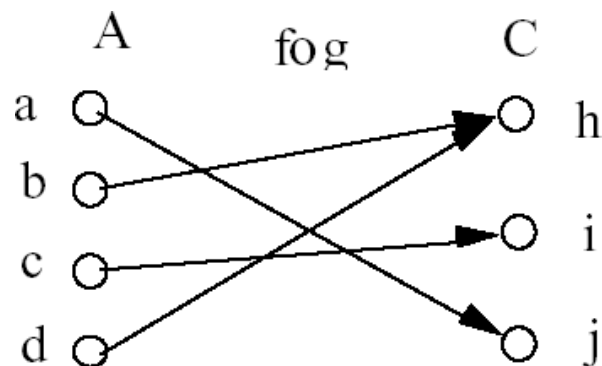
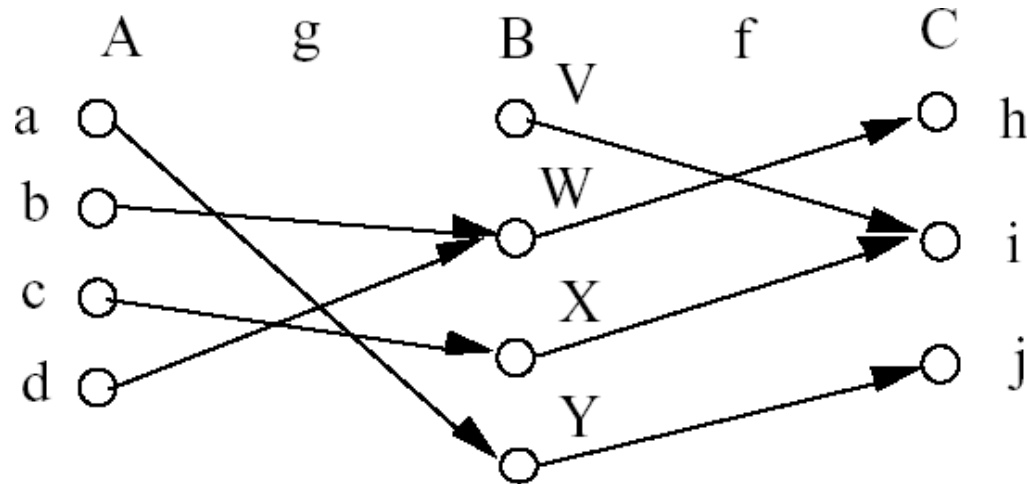
Function Composition Illustration

■ $g: A \rightarrow B, f: B \rightarrow C$



Function Composition: Example

■ $g: A \rightarrow B, f: B \rightarrow C$



Function Composition: Example

- Example 20: Let $g: \{a, b, c\} \rightarrow \{a, b, c\}$ such that $g(a) = b, g(b) = c, g(c) = a$.

Let $f: \{a, b, c\} \rightarrow \{1, 2, 3\}$ such that $f(a) = 3, f(b) = 2, f(c) = 1$.

What is the composition of f and g , and what is the composition of g and f ?

- $f \circ g: \{a, b, c\} \rightarrow \{1, 2, 3\}$ such that $(f \circ g)(a) = 2, (f \circ g)(b) = 1, (f \circ g)(c) = 3$.



- $g \circ f$ is not defined (why?)

Function Composition: Example

- If $f(x) = x^2$ and $g(x) = 2x + 1$, then what is the composition of f and g , and what is the composition of g and f ?

- $$\begin{aligned}(f \circ g)(x) &= f(g(x)) \\ &= f(2x+1) \\ &= (2x+1)^2\end{aligned}$$

- $$\begin{aligned}(g \circ f)(x) &= g(f(x)) \\ &= g(x^2) \\ &= 2x^2 + 1\end{aligned}$$

Note that $f \circ g \neq g \circ f$. ($4x^2+4x+1 \neq 2x^2+1$)

Images of Sets under Functions

- Given $f : A \rightarrow B$, and $S \subseteq A$,
- The **image** of S under f is simply the set of all images (under f) of the elements of S .

$$\begin{aligned} f(S) &= \{f(t) \mid t \in S\} \\ &= \{b \mid \exists t \in S: f(t) = b\}. \end{aligned}$$

- Note the range of f can be defined as simply the image (under f) of f 's domain.

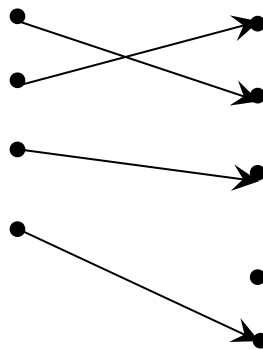
One-to-One Functions

- A function f is **one-to-one** ($1-1$), or **injective**, or an **injection**, iff $f(a) = f(b)$ implies that $a = b$ for all a and b in the domain of f (i.e. every element of its range has *only* 1 pre-image).
 - Formally, given $f : A \rightarrow B$,
“ f is injective”: $\forall a, b (f(a) = f(b) \rightarrow a = b)$ or equivalently $\forall a, b (a \neq b \rightarrow f(a) \neq f(b))$
- Only one element of the domain is mapped to any given one element of the range.
 - Domain & range have the same cardinality.
What about codomain?

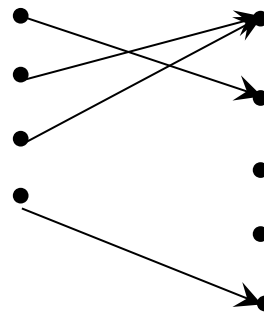


One-to-One Illustration

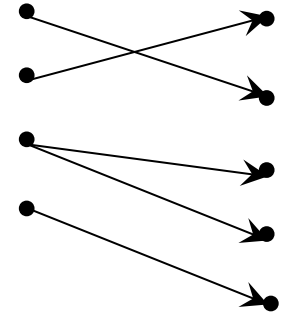
- Bipartite (2-part) graph representations of functions that are (or not) one-to-one:



One-to-one



Not one-to-one



Not even a function!

- Example 8:

Is the function $f : \{a, b, c, d\} \rightarrow \{1, 2, 3, 4, 5\}$ with $f(a) = 4$, $f(b) = 5$, $f(c) = 1$, and $f(d) = 3$ one-to-one?

- Example 9:

Let $f : \mathbf{Z} \rightarrow \mathbf{Z}$ such that $f(x) = x^2$. Is f one-to-one?

Sufficient Conditions for 1-1ness



- For functions f over numbers, we say:
 - f is **strictly** (or **monotonically**) **increasing**
iff $x > y \rightarrow f(x) > f(y)$ for all x, y in domain;
 - f is **strictly** (or **monotonically**) **decreasing**
iff $x > y \rightarrow f(x) < f(y)$ for all x, y in domain;
- If f is either strictly increasing or strictly decreasing, then f is one-to-one.
 - E.g. x^3

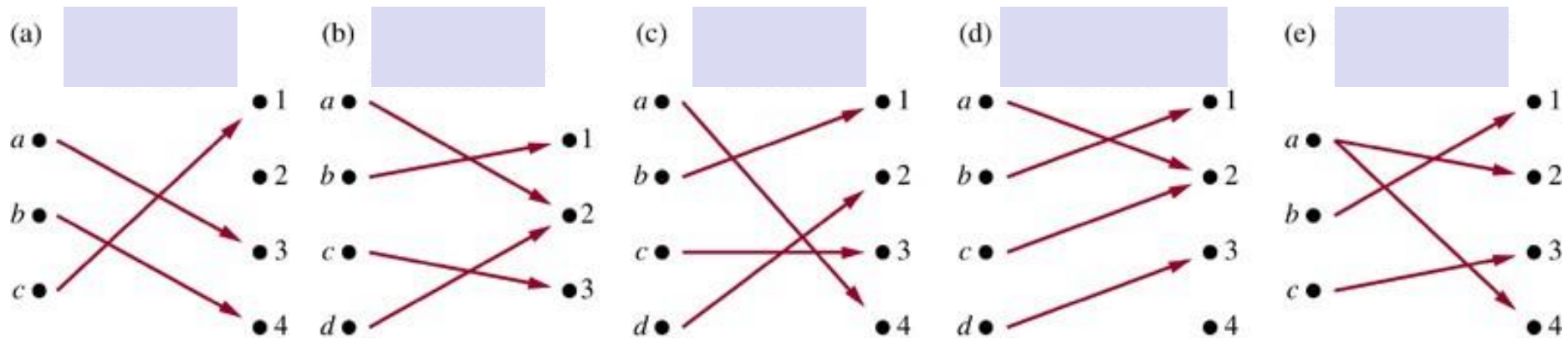
Onto (Surjective) Functions

- A function $f : A \rightarrow B$ is **onto** or **surjective** or a **surjection** iff for every element $b \in B$ there is an element $a \in A$ with $f(a) = b$ ($\forall b \in B, \exists a \in A: f(a) = b$) (i.e. its range is equal to its codomain).
- Think: An *onto* function maps the set A onto (over, covering) the *entirety* of the set B , not just over a piece of it.
- E.g., for domain & codomain $\mathbf{R}$, x^3 is onto, whereas x^2 isn't. (Why not?)

Illustration of Onto

- Some functions that are, or are not, *onto* their codomains:

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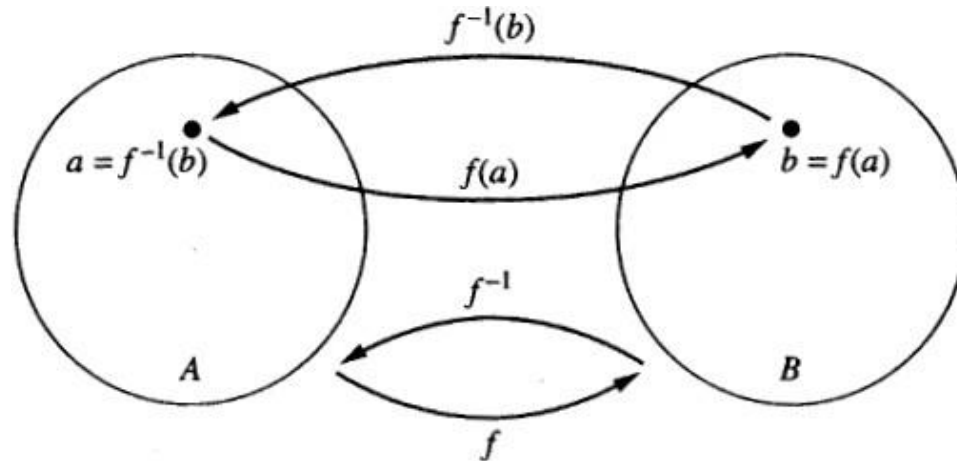
- Example 13: Is the function $f(x) = x + 1$ from the set of integers to the set of integers onto?

Bijections and Inverse Function

- A function f is said to be a ***one-to-one correspondence***, or a ***bijection***, or *reversible*, or *invertible*, iff it is both one-to-one and onto.
- Let $f : A \rightarrow B$ be a bijection.
The ***inverse function*** of f is the function that assigns to an element $b \in B$ the unique element $a \in A$ such that $f(a) = b$.
The inverse function of f is denoted by $f^{-1} : B \rightarrow A$.
Hence, $f^{-1}(b) = a$ when $f(a) = b$.

Inverse Function Illustration

- Let $f: A \rightarrow B$ be a bijection



- Example 16: Let $f: \{a, b, c\} \rightarrow \{1, 2, 3\}$ such that $f(a) = 2$, $f(b) = 3$, $f(c) = 1$. Is f invertible, and if it is, what is its inverse? Yes. $f^{-1}(1) = c$, $f^{-1}(2) = a$, $f^{-1}(3) = b$
- Example 18: Let f be the function from $\mathbf{R}$ to $\mathbf{R}$ with $f(x) = x^2$. Is f invertible? No. f is not a one-to-one function. So it's not invertible.

Mappings in Java

- A discrete function can be represented by a `Map` interface or `HashMap` class in Java programming language
 - `Map map<Integer,String>`
`= new HashMap<Integer,String>();`
 - Here, the domain is `Integer`, the codomain is `String`
- We can construct such a mapping by putting all pairs $\{a, f(a)\}$ into our `map`. (a is the *key*, $f(a)$ is the *value*.)
 - `map.put(2, "Jan");`
 - `for (Kid kid:kids) {map.put(kid.id, kid.name);}`
- If we put another pair with the same key, it will overwrite the previous pair – it's not a function! (May be a bug...)

Image, Range, Bijection in Java

- `Map.keys()` returns the **image**

- it's a Java Set!

- `map.values()` returns the **range**

- it's a Java Set!

- Is a map a bijection?

Iff the cardinalities of the **image** and **range** are the same:

- ```
if (map.keys().size()==map.values().size()) {
 System.out.println("map is a bijection");
}
```

# Inverse Function in Java

- Let's construct an inverse!
- Prepare the inverse function:
  - `Map inverse<String,Integer>`  
`= new HashMap<String,Integer>() ;`
  - Here, the domain is `String`, the codomain is `Integer`
- Go through all keys in `map` (all elements of the **image**) and put each pair `{value,key}` into `inverse`:
  - ```
for (Integer id:map.keys()) {  
    String name = map.get(id);  
    inverse.put(id:name,id);  
}
```